

Windows Server 2025 Monitoring Lab using Prometheus and Grafana

Built a Windows Server 2025 infrastructure monitoring lab using Prometheus, Windows Exporter, and Grafana. Configured real-time dashboards and alerts for CPU, memory, disk, network, service status, and server uptime.

Setup:

VM 1: Domain Controller

VM 2: File Server

VM 3: Monitoring Server

VM4: Windows 11 client

What You'll Monitor

-Domain Controller (DC01)

- CPU Usage
- Memory Usage
- Disk Space
- Server Uptime
- Network Traffic
- DNS Service Status
- AD DS Service Status

-File Server (FILE01)

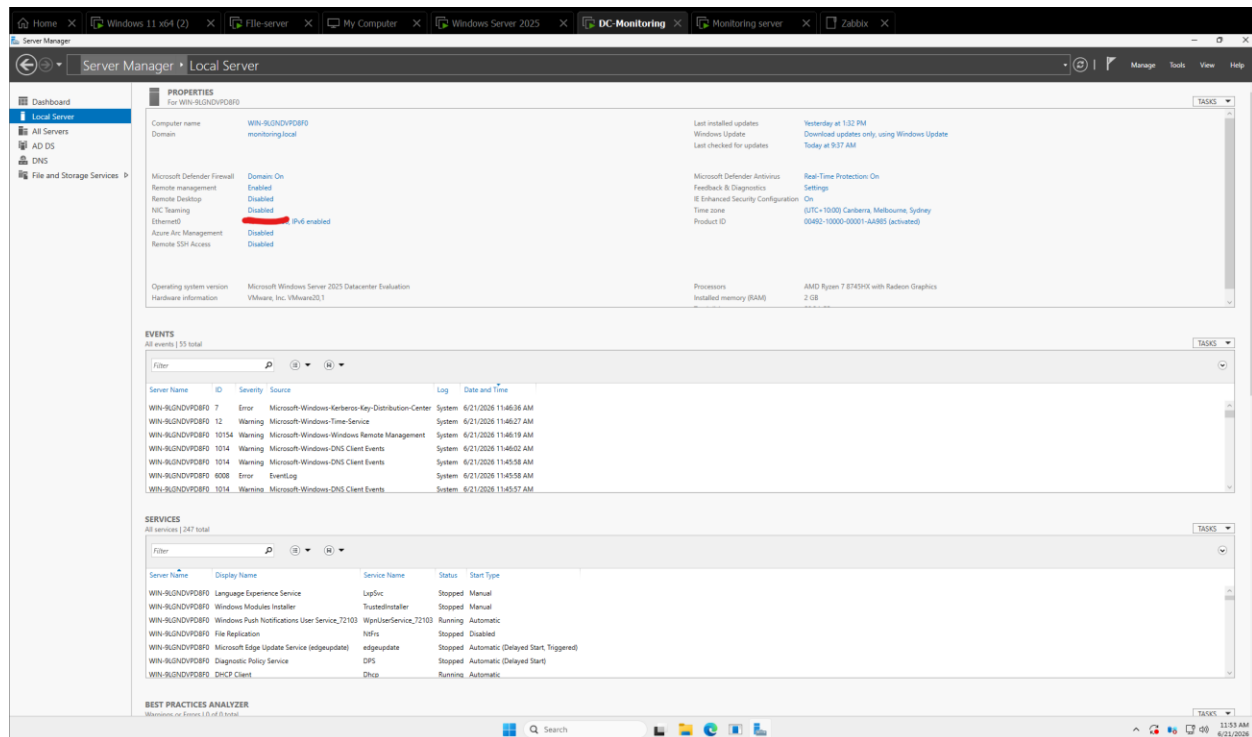
- CPU Usage
- Memory Usage
- Disk Space
- Network Traffic
- SMB Service Status
- Shared Folder Availability
- File Server Uptime

Now your lab demonstrates:

- Active Directory
- DNS
- Group Management
- File Services
- NTFS Permissions
- Infrastructure Monitoring
- Grafana Dashboards
- Prometheus Monitoring

VM 1: Domain Controller

Virtual machine is created and promoted to domain controller and has a domain name monitoring.local.



Join other VMs into the domain

Create OUs:

Corporate

- Users
- Groups
- Servers

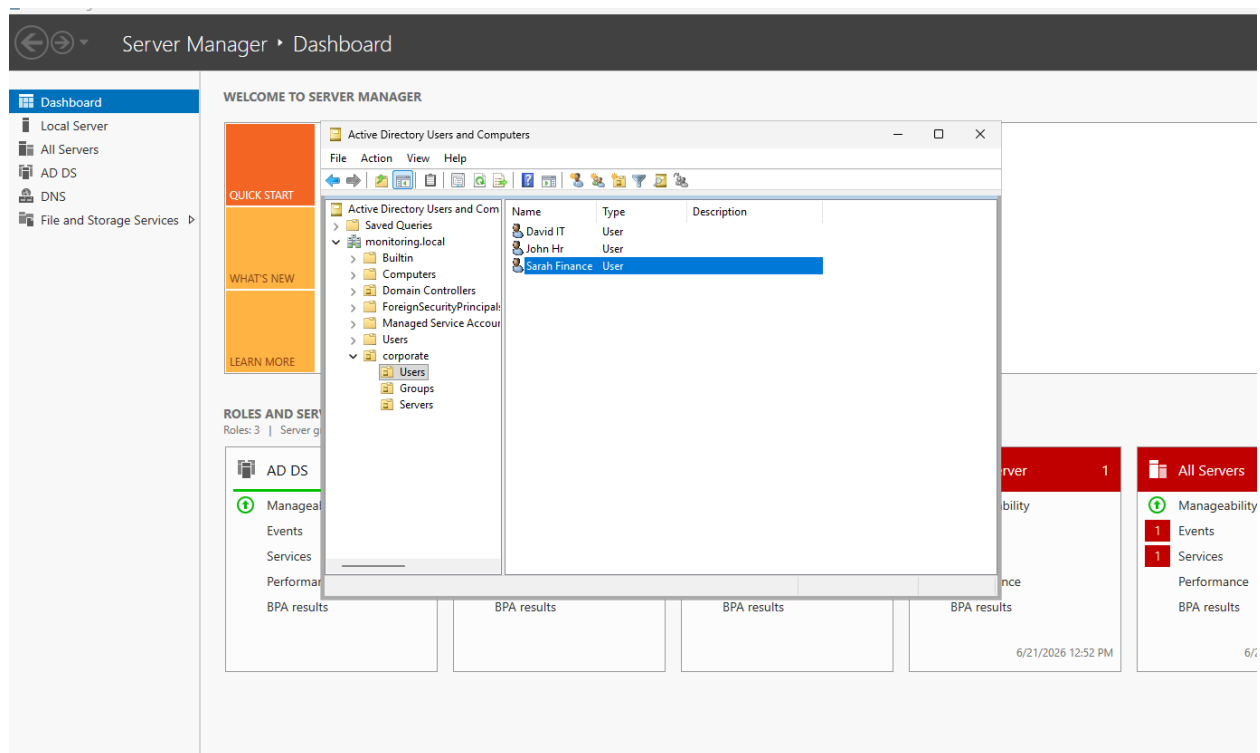
Create Users

Inside Users OU:

John.HR

David.IT

Sarah.Finance



Inside Groups OU: Add users to the group

Local server

Users

to manage

Group

Connect to cloud services

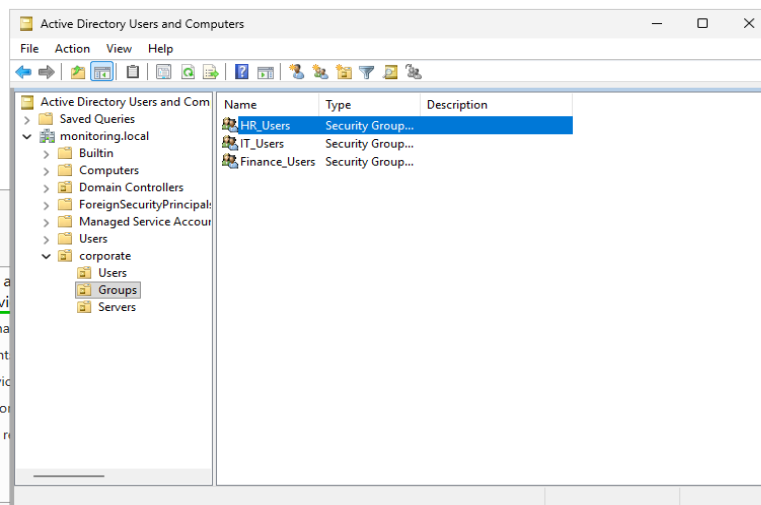
1

Manageability

Performance

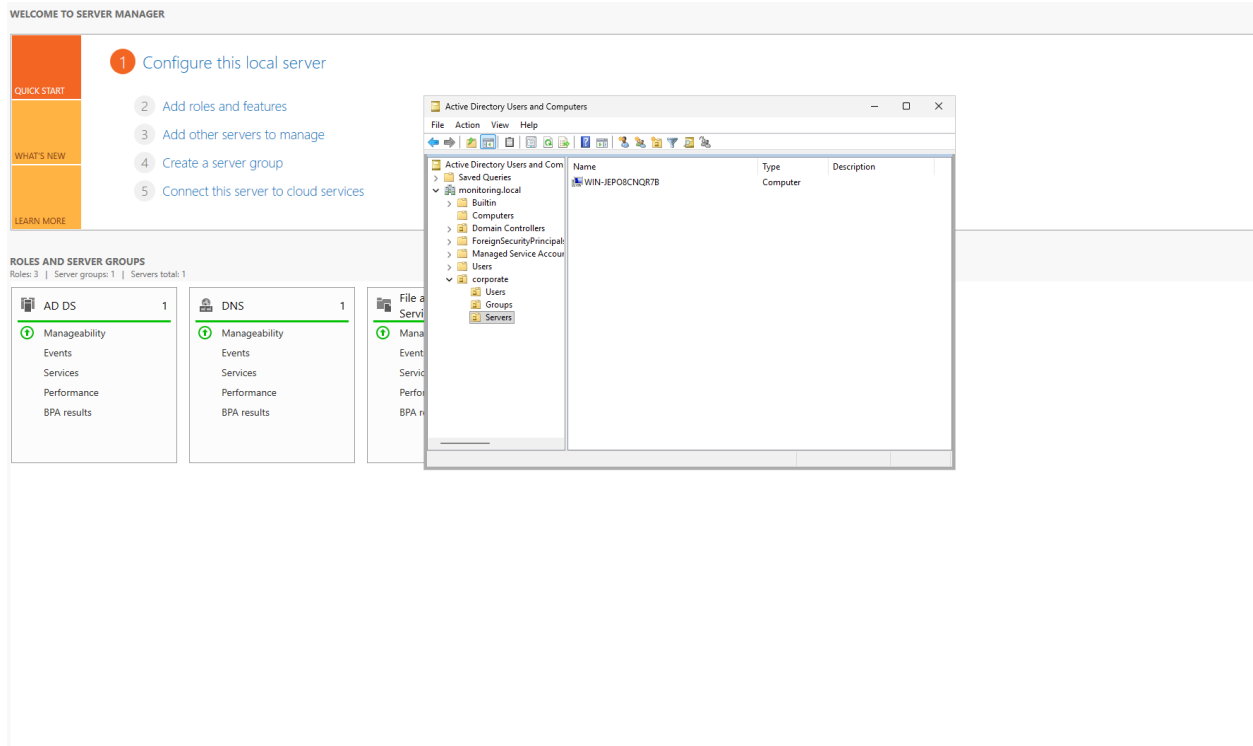
Services

BPA results

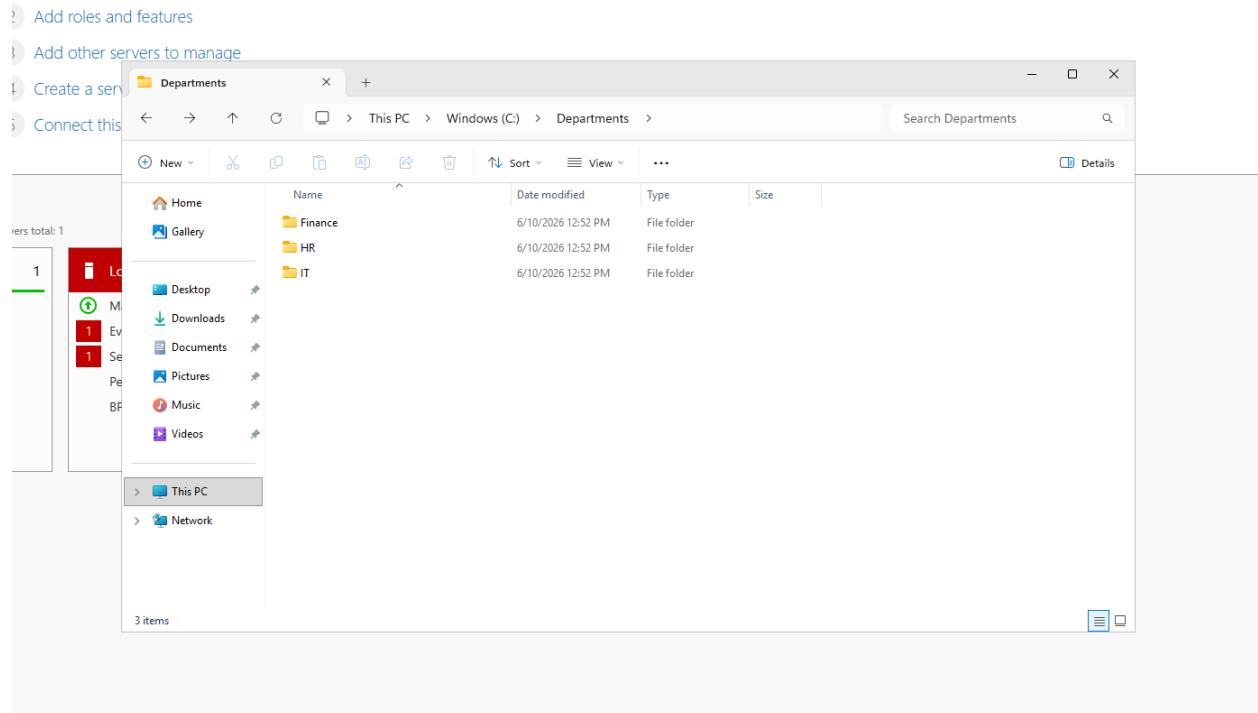


Move Server Object

On DC: Move FILE01 into: Servers OU



Create Department Folders



Create SMB Shares: Server Manager → File and Storage Services → Shares
Click: Tasks → New Share

Create HR Share

(Or)

Right-click folder → **Properties** → **Sharing** → **Advanced Sharing** → **Permissions**

Configure NTFS Permissions:

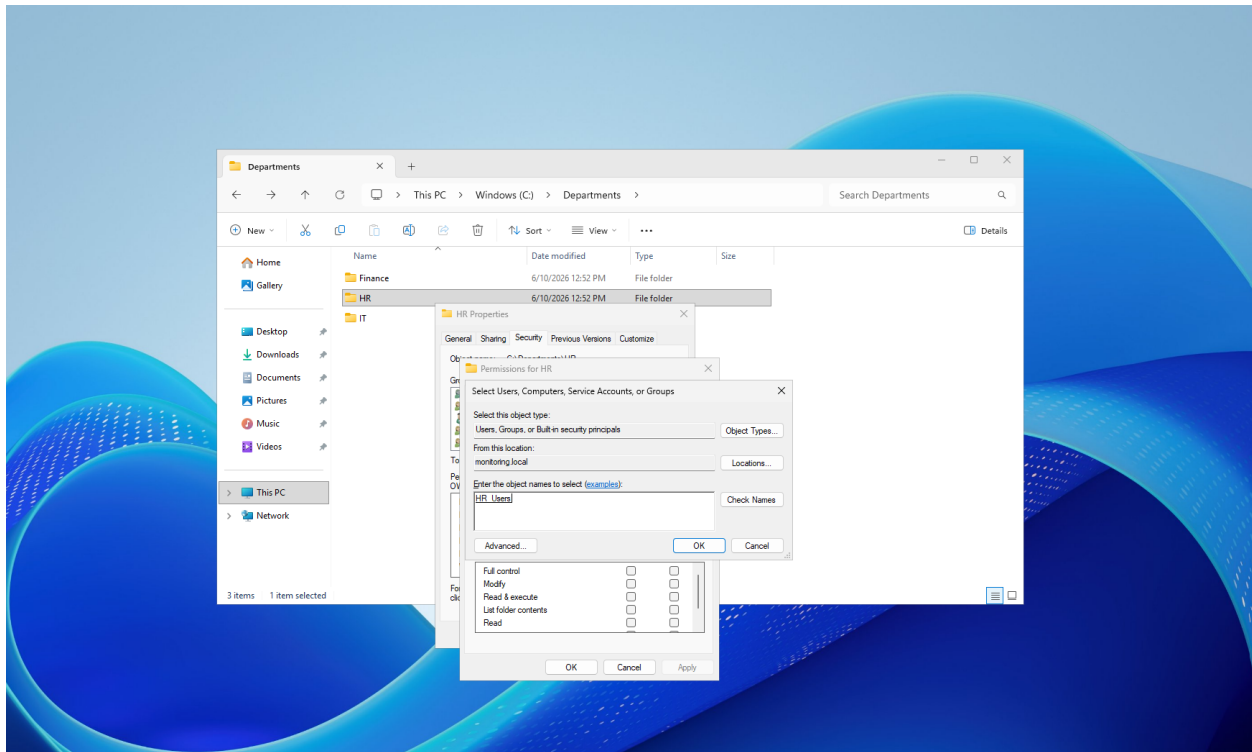
Go to:

Properties → **Security**

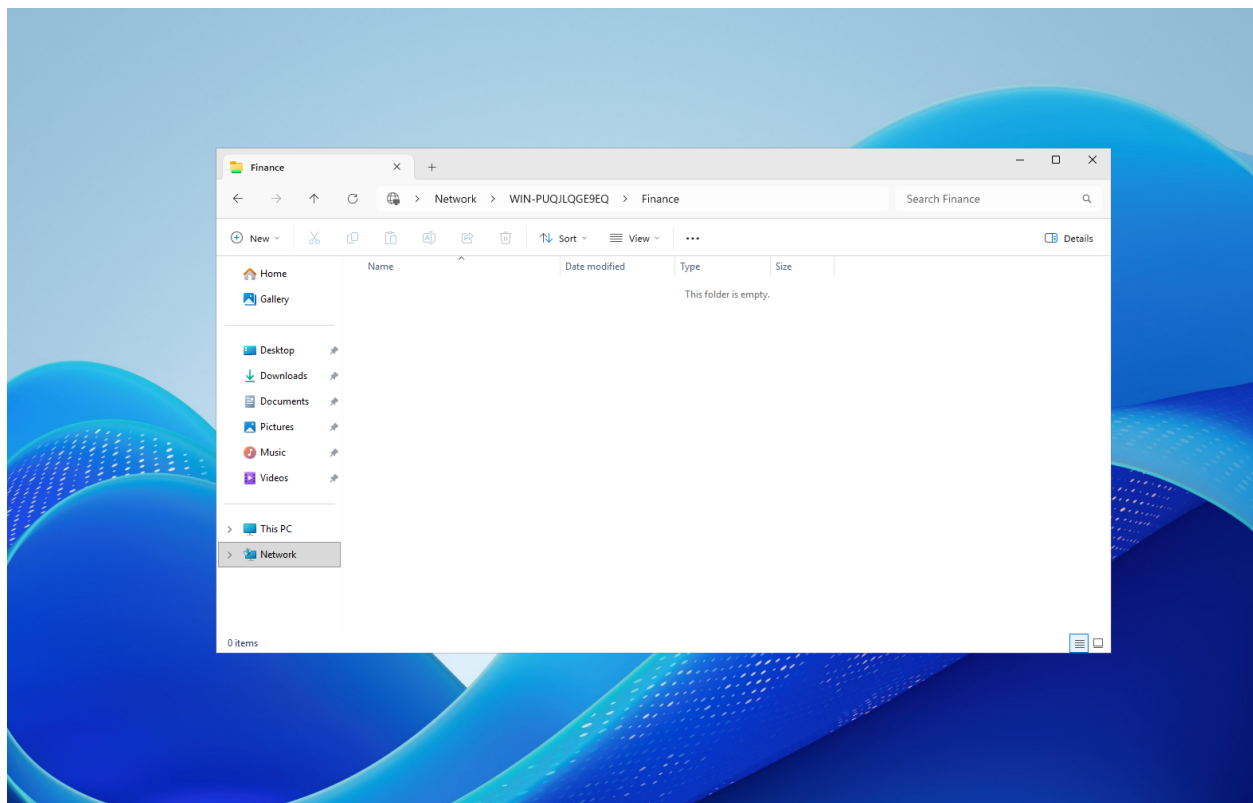
For HR folder:

Add:

HR_Users

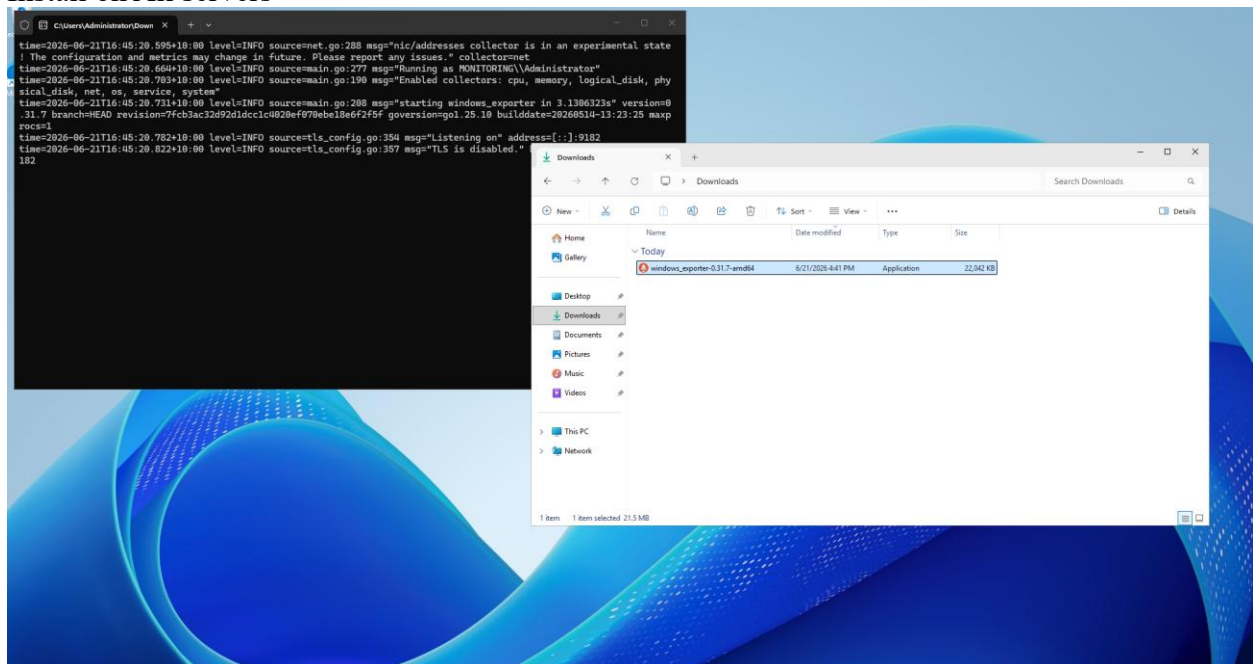


I can access the file share on monitoring server using Sarah user



Phase 7 – Install Windows Exporter

Install on All servers

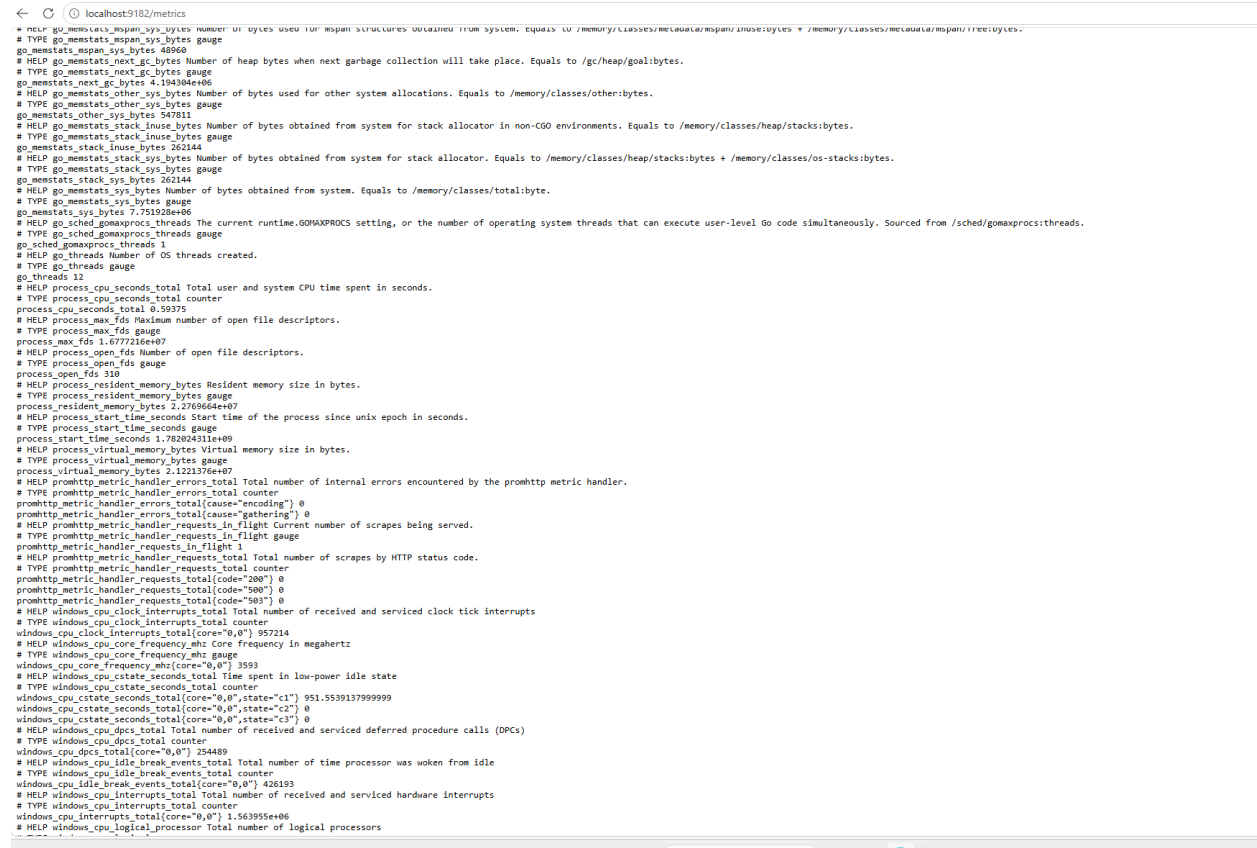


Verify Metrics

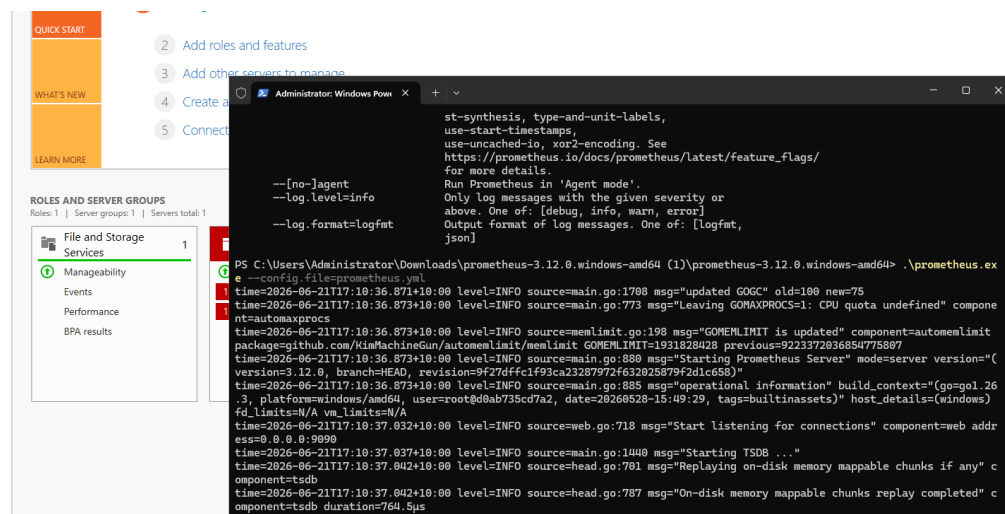
Open:

<http://localhost:9182/metrics>

You should see thousands of metrics.

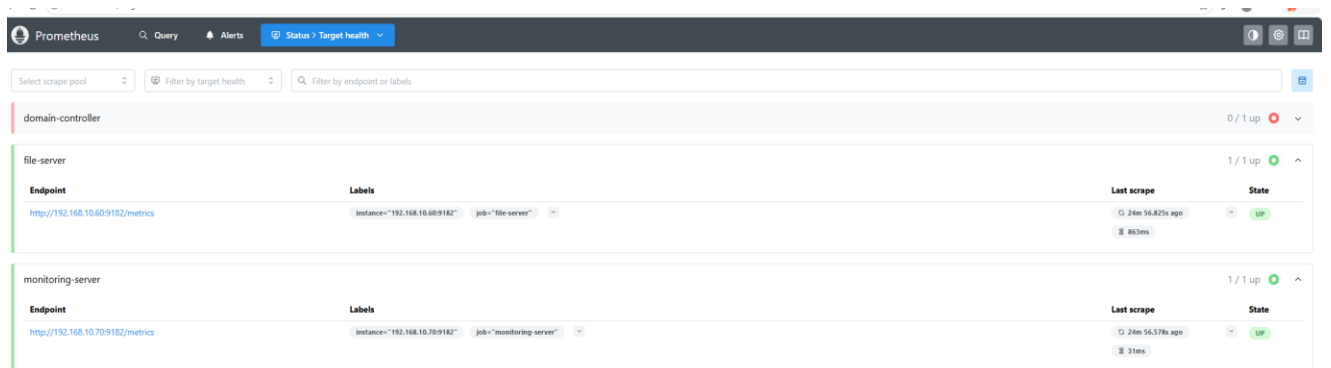
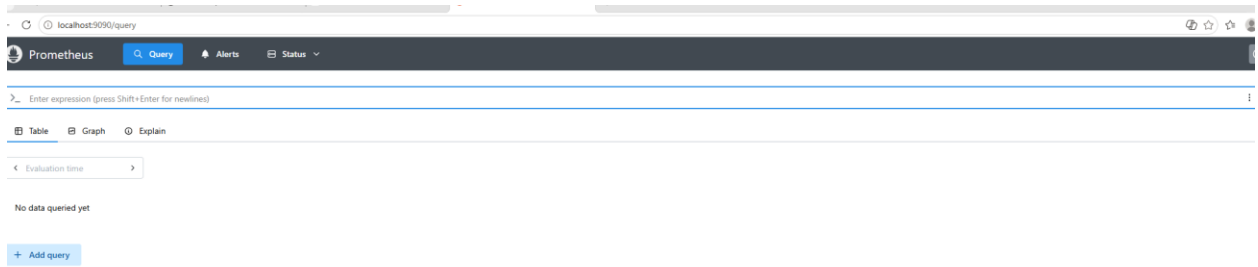


Install Prometheus on monitoring server and start Prometheus:

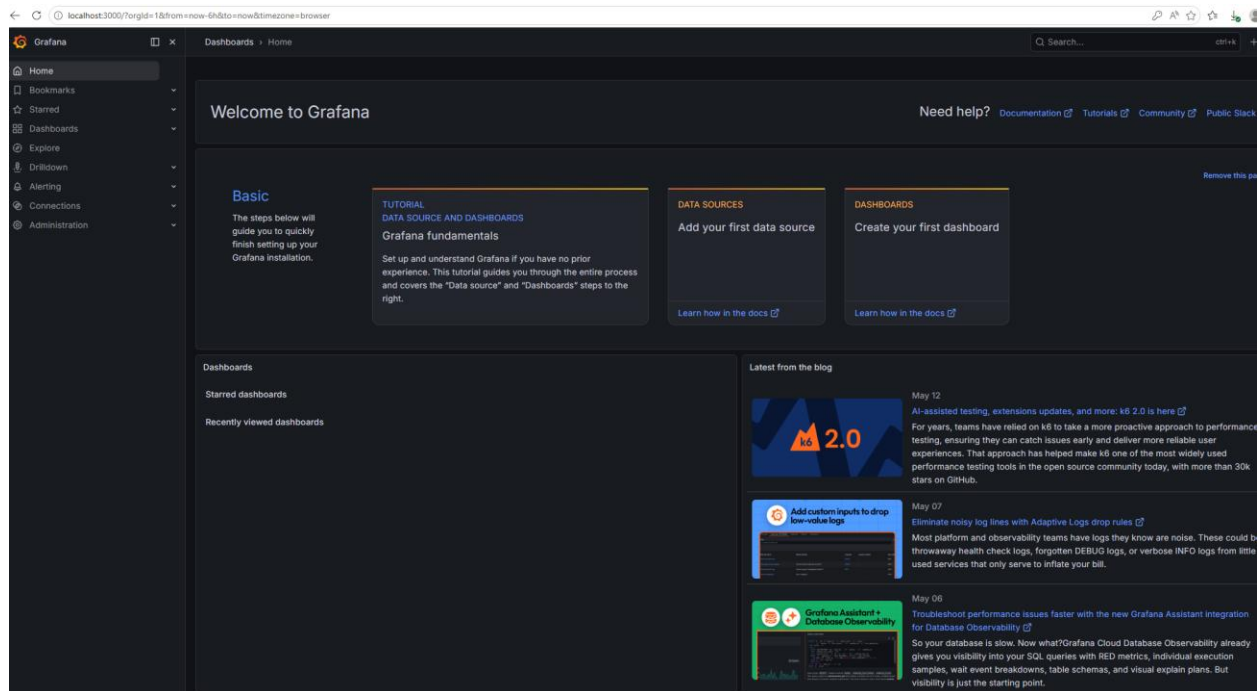


Open:

http://localhost:9090
Go: Status and Targets

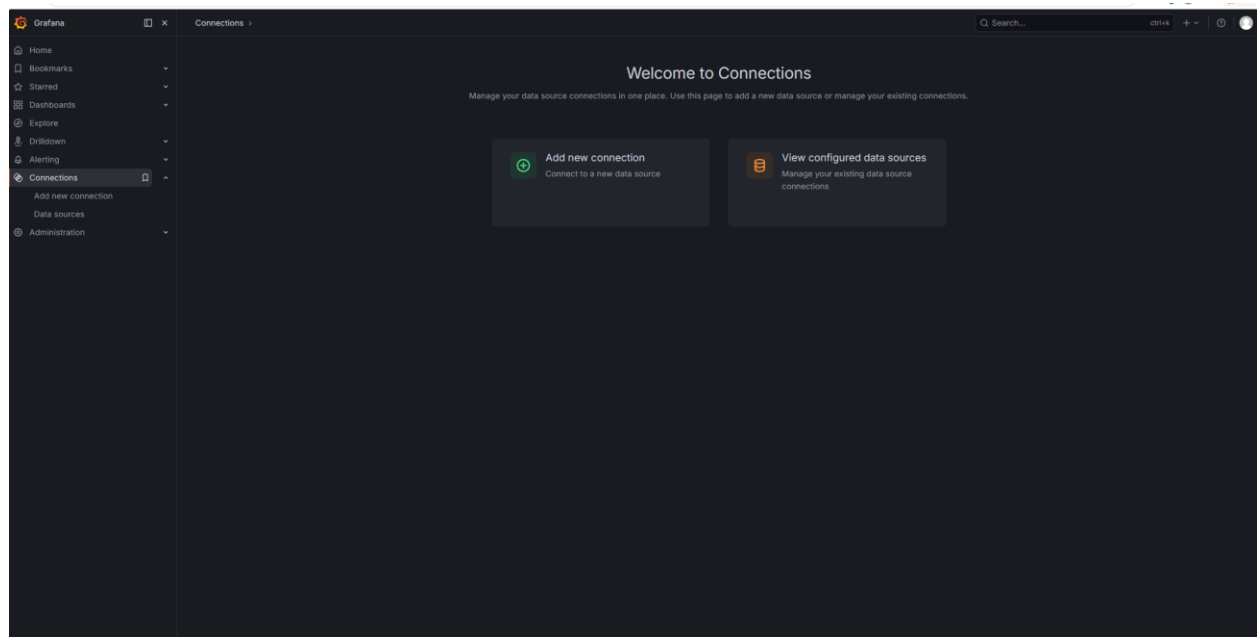


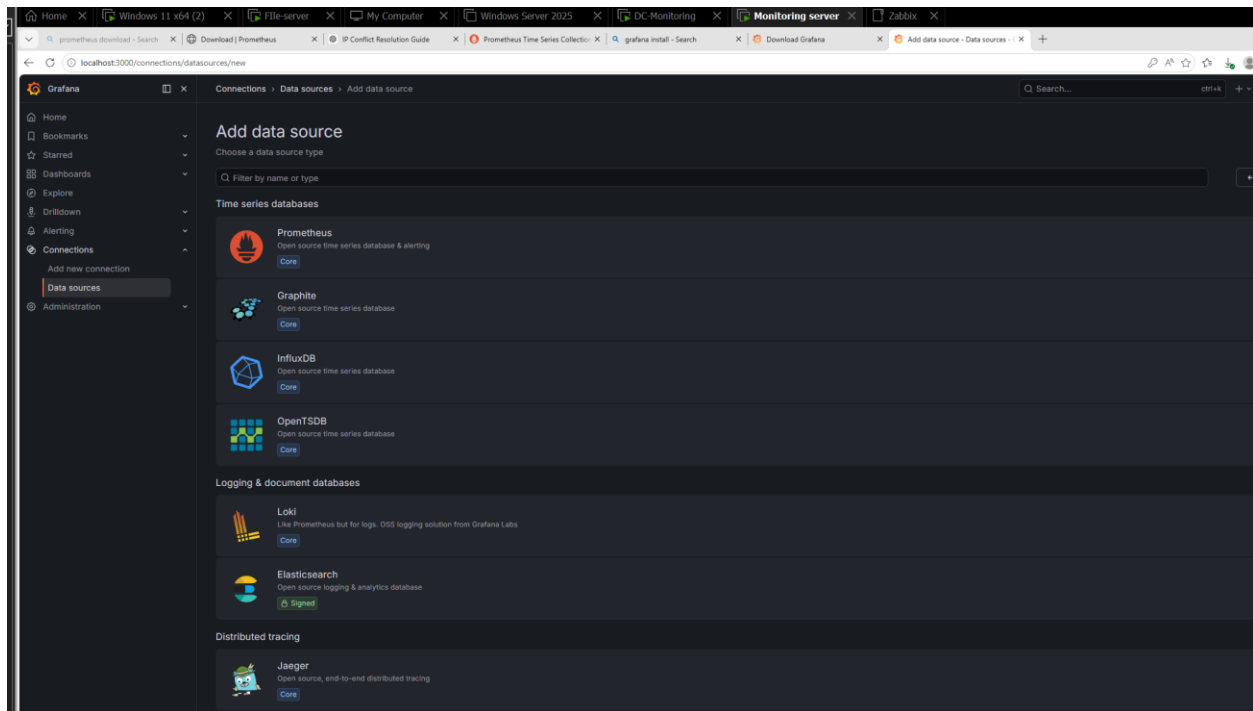
Install Grafana and open in the browser



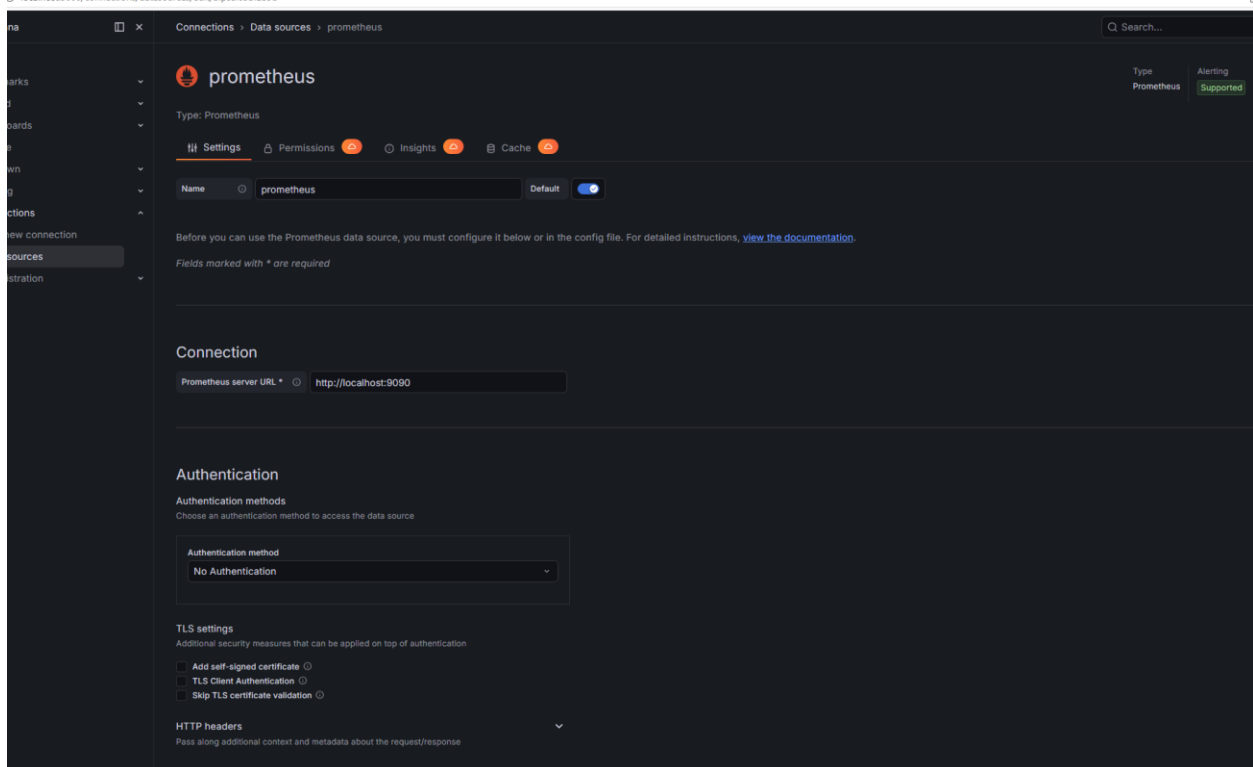
Add Prometheus

Go:->Connections->Data Sources->Prometheus





Enter the URL

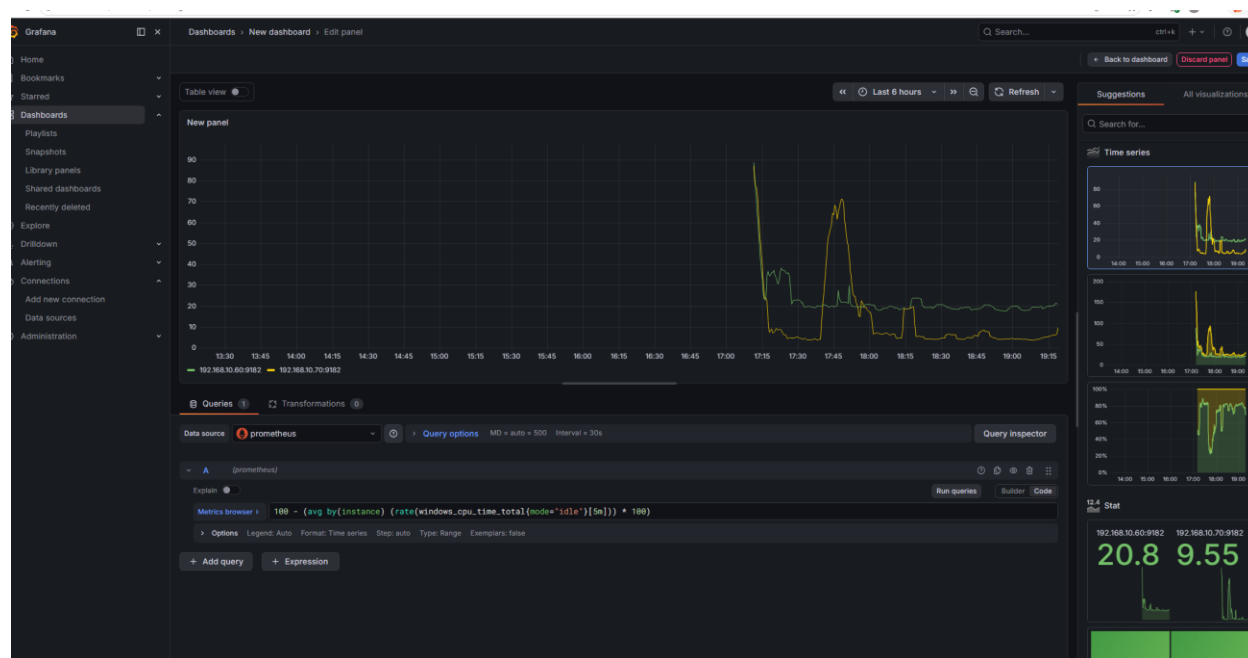


And save

To create Dashboard:

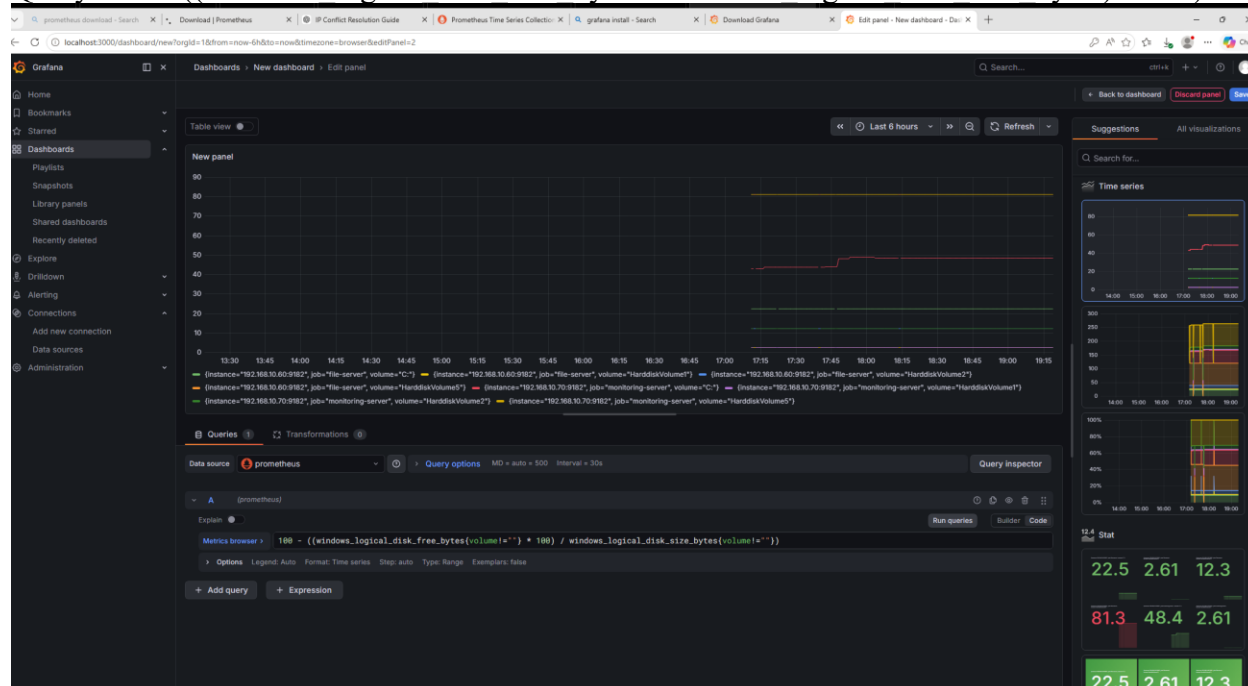
CPU Usage:

$100 - (\text{avg by(instance)} (\text{rate}(\text{windows_cpu_time_total}\{\text{mode}=\text{"idle"}\}[5\text{m}])) * 100)$



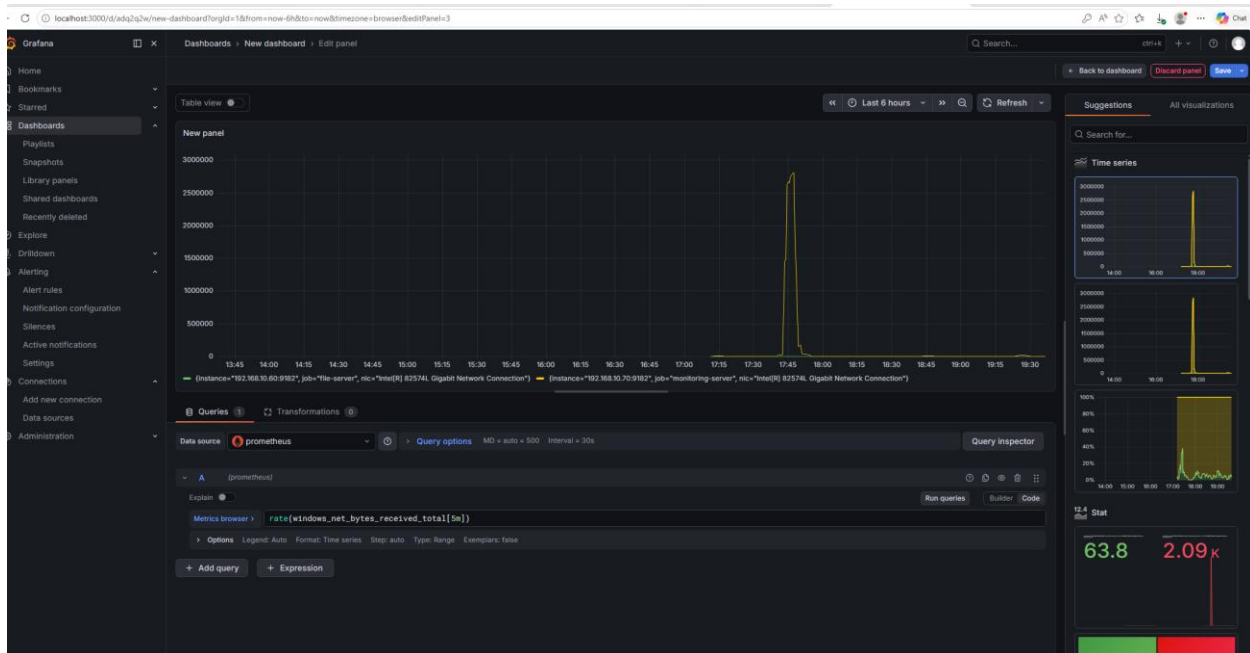
Disk Usage:

Query: $100 - ((\text{windows_logical_disk_free_bytes} / \text{windows_logical_disk_size_bytes}) * 100)$



Network traffic:

Query: $\text{rate}(\text{windows_net_bytes_received_total}[5\text{m}])$

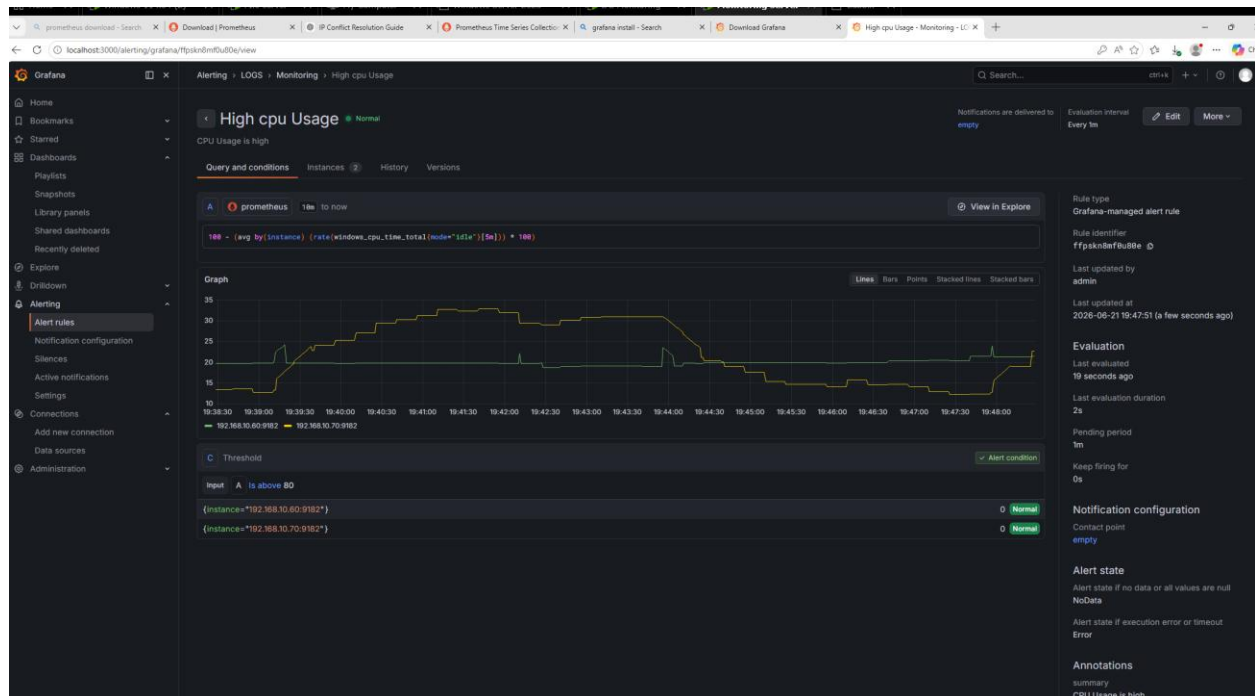


To create Alerts:

Alerting→ Alert Rules→ Create Alert Rule

The screenshot shows the 'New alert rule' configuration page in Grafana. The page is divided into four sections:

- 1. Enter alert rule name**: The 'Name' field is set to 'High cpu Usage'.
- 2. Define query and alert condition**: The 'Query' field is set to 'rate(window_net_bytes_received_total[5m])'. The 'Alert condition' is set to 'WHEN QUERY IS ABOVE 0'.
- 3. Add folder and labels**: The 'Folder' is set to 'Alerting'. The 'Labels' field is empty.
- 4. Set evaluation behavior**: The 'Evaluation interval' is set to '1m'.



You should see:

High CPU Usage

Status: Normal

(not Firing, because your threshold is 80 and current CPU is only ~20–30%).

Summary:

- Built a multi-server Windows Server 2025 environment consisting of Active Directory, DNS, and File Services.
- Configured role-based access control using security groups and NTFS permissions for departmental file shares.
- Implemented centralized monitoring using Prometheus, Windows Exporter, and Grafana to track server performance and availability.
- Developed dashboards and alerts for CPU, memory, disk utilization, uptime, network traffic, and Windows services.